

Resource Estimation for Delayed Choice Quantum Entanglement Based Sneakernet Networks Using Neutral Atom qLDPC Memories

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Quantum Entanglement is a vital phenomenon required for realizing secure quantum networks, so much that distributed entanglement can be re-imagined as a commodity which can be traded to enable and maintain these networks. We explore the idea of commercializing entanglement-based cryptography and future applications where advanced quantum memory systems support less advanced users. We design a sneakernet-based quantum communication network with a central party connecting the users through delayed-choice quantum entanglement swapping, using quantum Low-Density-Parity-Check (qLDPC) encoded qubits on neutral atoms. Our analysis compares this approach with traditional surface codes, demonstrating that qLDPC codes offer superior scaling in terms of resource efficiency and logical qubit count. We show that with near-term attainable patch sizes, one can attain medium-to-high fidelity correlations, motivating further research towards the long-term realization of large-scale commercial quantum networks.

I. INTRODUCTION

The second Quantum Revolution refers to a future technological period when all the features of quantum mechanics are harnessed for information processing devices, expanding beyond the limited application of quantum mechanics in current computers, communication systems, and sensors [1–5]. The ultimate achievement of this revolution is embodied by quantum computers that are capable of operating on a vast scale, with the ability to tolerate faults and fix errors. These computers would need millions or more logical qubits [6] to tackle complex cryptographic tasks. While the creation of such devices remains in its early stages, there are compelling grounds to anticipate that scalable quantum computers will become tangible at some point within this century [7].

While there is reason to be hopeful about the progress of large-scale quantum computers, it remains uncertain if quantum computers for everyday consumers, such as a quantum smartphone, will ever become feasible. This refers to a compact and powerful computational device that is connected to a quantum network, possibly on a global scale. Possible constraints that could prevent the development of quantum smartphones include the requirement for extremely low temperatures to cool down qubits [8], or the necessity for optical communication channels [9] instead of microwave ones to connect to the quantum internet, the vulnerability of qubits to mechanical vibrations and the impact of the Earth’s magnetic field [10]. Suppose we accept the possibility of large-scale quantum computers and a quantum internet. In that case, it is logical to enquire how a customer, who has no access to a quantum memory, but only has access to conventional gear such as classical detectors, classical memory, and classical computing, can participate in and

perform complex quantum protocols.

One way is to do this by the use of distributed quantum entanglement [11]. Quantum entanglement is a pivotal phenomenon that forms the basis of various protocols in quantum communications and quantum computing [12–17]. Due to it being a major primitive for most quantum tasks, it would be natural to consider quantum entanglement as a tradeable commodity, such that users can purchase it to establish connections with other users in the network to perform quantum tasks such as quantum key distribution, distributed quantum computing, quantum authentication, etc. Commoditizing entanglement is an essential step toward large-scale commercial quantum networks.

To better demonstrate the above-mentioned ideas of entanglement commercialization and quantum access to common users with classical hardware, we design and analyse an entanglement one-time-pad QKD [18] based quantum network model networked using delayed-choice entanglement swapping [19]. In this study, we demonstrate that delayed-choice entanglement swapping enables customers, who only own classical hardware, to use the offerings of a hypothetical future quantum memory company for the purpose of generating secure quantum keys. These keys are based on the E91 protocol [20]. Our design employs a central party Charlie, who generates Bell-pairs, loads the halves separately onto different qLDPC codes, and distributes one set of the halves via sneakernet [21, 22] to multiple endpoints where it is immediately measured upon arrival while keeping the other set of halves to himself on which he later performs Bell measurements to generate correlations between the endpoint measurements of any pair of users. This enables the users to execute this task in a versatile manner by pre-purchasing their key bits prior to selecting their communication partners. This protocol is beneficial for creating secure connections between networks of devices

that have minimal hardware needs but can interact with a single, advanced quantum node or several nodes.

We design our protocol on quantum Hyper Graph Product (HGP) codes [23], which are a class of Low-Density Parity-Check (qLDPC) codes [24] for quantum error correction [25], implemented on a neutral atom architecture [26, 27]. These codes offer significant advantages over traditional surface codes [28] in terms of encoding rate and resource efficiency [27, 29]. Our analysis provides a detailed comparison between qLDPC and surface codes, demonstrating that qLDPC codes allow us to achieve higher logical qubit counts with fewer physical qubits as they have constant encoding rates regardless of patch size. This property potentially reduces the hardware requirements and improves the scalability of our proposed system. We have selected neutral atoms as our physical platform, as their ability to shuttle and rearrange within the lattice enables selective qubit unloading and efficient syndrome extraction [27]. This capability also allows us to use only one set of ancilla qubits and a single surface code patch, positioned adjacent to the qLDPC code block, for both loading and unloading operations via teleportation [27].

The foundation of our study is based on the research conducted on consumer-level quantum key distribution [30–32], as well as the research on quantum sneakernet [21], which explores the spread of entanglement across long distances with high bandwidth and low latency. While our primary focus is on secure communication protocols, it is feasible to adapt our methodology for many applications such as quantum voting [33], clock synchronization [34], and reference frame alignment [35].

This study is organized as follows: In section II, we briefly describe the phenomenon of delayed choice entanglement swapping. In section III, we describe our design. Section IV derives the closed-form mathematical expressions necessary for our resource estimations. In section V we discuss our findings, and finally conclude our study in section VI. Link to the python code of our analysis is here: <https://shorturl1.at/QZ9v7>.

II. DELAYED CHOICE ENTANGLEMENT SWAPPING

To describe our protocol, we start by providing a concise overview of the initial delayed-choice entanglement-swapping technique, as explained by Peres [19]. In accordance with Peres, we examine the most basic iteration of this protocol. This protocol aims to establish entanglement and eventually generate correlated bits between two parties, Alice and Bob, who choose a third party, Charlie, to mediate the entanglement generation. Consider three individuals, namely Alice, Bob, and Charlie, as seen in the circuit diagram in Figure 1. Two Bell Pairs q_0q_1 and q_2q_3 are present such that q_0 is in the custody of Alice, q_3 with Bob, and q_1 and q_2 with Charlie. Qubits q_0 of Alice and q_3 of Bob get entangled when Charlie

performs a joint Bell-measurement on his qubits q_1 and q_2 . However, the specific relationship between Alice and Bob’s qubits cannot be known until Charlie reveals the outcome of his Bell state measurement. Alice and Bob’s photons are perfectly entangled, disregarding any errors or decoherence. This exact entanglement allows them to be used for various entanglement-based protocols, such as cryptography using the E91 protocol [20].

Alternatively, Charlie has the option to measure his photons separately. By adopting this approach, there would be a complete absence of entanglement between the photons belonging to Alice and Bob. Consequently, any correlations seen would be purely coincidental and not indicative of any underlying connection. If Charlie attempted to conceal his independent measurements, Alice and Bob might easily expose the deceit by using the conventional method of conducting measurements and announcing certain findings subsequent to receiving Charlie’s ‘correction’ signals.

Peres’ idea goes beyond entanglement swapping by recognizing that Alice, Bob, and Charlie might potentially be space-like separated. This suggests that one can discover a frame of reference when the sequence of qubit measurement is modified, as shown by the Gisin group [37]. In the preceding discourse, we used the assumption that Charlie conducted his measurement before Alice and Bob did. Alternatively, it is possible for Alice and Bob to conduct their measurements prior to Charlie (Figure 1(b)). In this scenario, the outcome of their measurements itself is not affected by whether or not Charlie performed either the Bell state or independent measurement, but the degree of correlation between Alice and Bob’s measured bits is dependent upon the kind of measurement performed by Charlie. This serves as the foundation for the delayed-choice entanglement-swapping protocol and is crucial for the advancement of our ideas. In the delayed-choice entanglement swapping technique, Alice and Bob perform measurements on their qubits and record the outcomes using classical storage. At a later point, Charlie has a decision to either measure his qubits separately or do a Bell state measurement. The measurement outputs are either associated in a known manner or wholly uncorrelated, depending on the measurement option chosen by Charlie and the random basis choice of Alice and Bob.

III. SETUP

We now assume that Charlie is replaced by a company Charlie Inc. whose business is to store and distribute entangled states. This requires Charlie Inc. to have large-scale long-lived quantum memory and a means of distributing entangled states. Our protocol is detailed in Figure 2. and is also described concisely in pseudocode 1. Charlie Inc., acting as a central node, first prepares Bell pairs in surface codes and teleports each half into separate qLDPC code blocks via a qLDPC (HGP) ancilla.

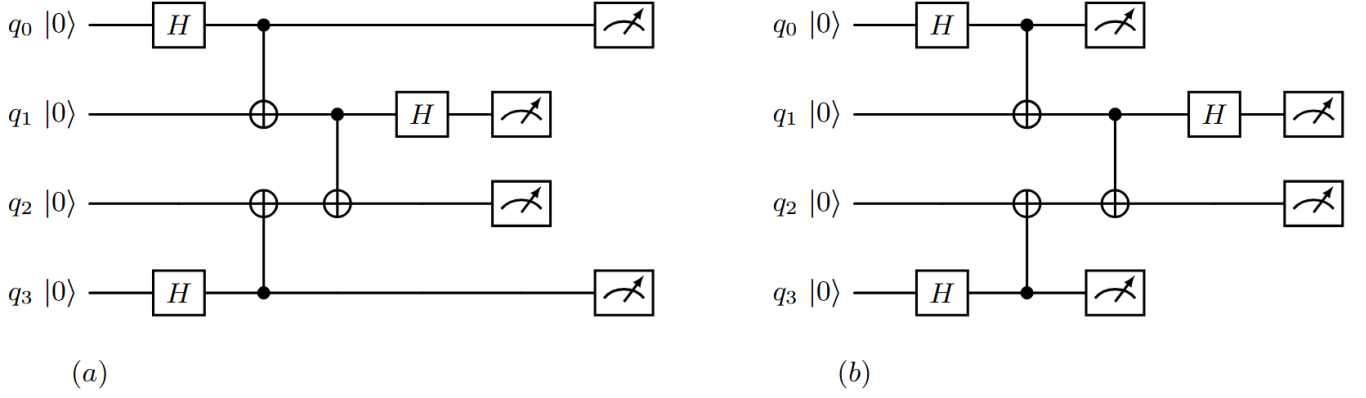


Figure 1. Circuit diagram of quantum entanglement swapping. In (a), the conventional entanglement swapping [36] occurs, with Charlie performing the Bell measurement before Alice and Bob does. In (b), the order of measurement is reversed, with Alice and Bob measuring their qubits before Charlie. In both the circuits, all the qubits are starting at $|0\rangle$ state. These two circuits are equivalent as the two sets of measurements are parallel, non-overlapping, and independent of each other.

Algorithm 1 Delayed Choice Quantum Network Protocol

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1: for all end-users do
2:   Charlie prepares Bell pairs encoded in surface codes
3:   Charlie loads each half of the Bell pairs into separate sets of qLDPC code blocks via teleportation
4:   Charlie stores one set of qLDPC blocks and transports the other set to destination
5:   End-user puts in a purchase order for  $n_{bits}$  number of bits as per their requirement
6:   for  $i$  from 1 to  $n_{bits}$  do
7:     qATM at the destination unloads a logical qubit from a qLDPC block onto a surface code
8:     qATM measures the logical surface code qubit to get a bit
9:     qATM transfers the bit onto the user's smartphone
10:    Bit stored in the end-user's smartphone for later use.
11:   end for
12: end for
13: Publish measurement bases
14: for users Alice and Bob wishing to connect do
15:   Alice and Bob decide to correlate  $n_{key}$  number of bits
16:   for  $i$  from 1 to  $n_{key}$  do
17:     Charlie unloads two corresponding qLDPC qubits onto two surface code patches
18:     Charlie performs logical Bell measurement on surface code patches
19:     Charlie broadcasts the measurement result to Alice and Bob, resulting in a correlated key bit
20:   end for
21:   Alice and Bob perform standard reconciliation techniques and get the final secure key
22: end for

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This is done because direct computations and measurements for qLDPC codes are not yet well defined at the time of this writing. Charlie Inc. then keeps one set of neutral-atom encoded qLDPC blocks and sends the other set to an end-user via physical transport, like a train or a truck. This process is done for several such end-users at different destinations. This is called the "sneakernet" method. At the user's end, the user purchases a certain number of bits based on their needs. At the user's end, a quantum ATM (qATM) "unloads" the incoming qLDPC blocks by sequentially teleporting the constituent qubits onto a surface code, followed by measuring the surface code randomly in X or Z bases, and commercially selling the resulting classical bits to users, which is in turn stored in a device belonging to the user, such as a smart-

phone. For our analysis, we assume that the process of transferring classical bits onto the end-users' classical devices is secure. The measurement bases are then made public.

Among several end-users in the network, let's say two of them, namely Alice and Bob, want to connect at a later time. They decide upon how many bits they need to correlate for their requirement. Charlie Inc. then converts the corresponding logical qubits in the qLDPC blocks on his end into surface code patches and performs Bell measurements on those patches, thereby correlating Alice and Bob's bits. Finally, Alice and Bob use their correlated bits to generate a draft encryption key. Depending on their security requirement, they can enhance the key's security using CHSH tests [38] and privacy am-